COVID-19 GLOBAL IMPACT DASHBOARD

Day 5 Final Project Report | Analysis, Interactive Dashboard & Executive Summary

1. Project Objective

Analyze publicly available COVID-19 time-series data to explore reported cases, deaths, vaccination coverage, and differences between locations. The project combines data preparation, trend analysis, population-normalized comparisons, and an interactive Power BI dashboard.

2. Dataset Overview

Item	Description
Source	Our World in Data COVID-19 dataset (owid-covid-data.csv)
Observed dataset size	395,311 rows × 72 columns before additional project calculations
Observed date range	2020-01-01 to 2024-08-14
Key fields	location, date, total_cases, new_cases, total_deaths, new_deaths, population and vaccination metrics

3. Tools and Methodology

- Python and Pandas for loading, inspecting, cleaning and transforming time-series data.
- Validation of date/location fields, duplicate location-date records, missing values and numeric data types.
- Derived metrics may include 7-day rolling averages, cases/deaths per million and reported case-fatality ratio.
- Matplotlib/Seaborn for exploratory charts; Power BI Desktop for interactive report pages and filters.
- Dashboard pages: Global Overview and Vaccination Analysis, with KPI cards, trend charts, country comparisons and slicers where corresponding fields are available.

4. Analysis and Visuals

- Global daily reported cases trend.
- Global daily reported deaths trend.
- Country comparison using reported cumulative cases per million.
- Vaccination coverage and total vaccinations per 100 people over time and by country.

5. Key Findings — Add Verified Final Values

To avoid inventing results, enter the final peak dates/values and country comparisons from the completed notebook and Power BI visuals before publishing.

- Cases trend: [insert observed peak date and value].
- Deaths trend: [insert observed peak date and value].
- Country comparison: [insert location, reported cases per million, and record date].
- Vaccination: [insert location, vaccination coverage, and record date].

6. Limitations

- Reported cases and deaths do not include every infection or death; testing and reporting differ across places and time.
- Vaccination fields have substantial missingness; comparisons should use available records and identify record dates.
- Per-million metrics help account for population size but do not eliminate surveillance and reporting differences.
- Case-fatality ratio is reported deaths divided by reported cases, not an individual's actual probability of death.
- Trend relationships alone do not establish that vaccination caused a change in cases or deaths.

7. Deliverables Checklist

Deliverable	Expected output
Cleaned data	CSV and documented assumptions
Charts	Cases, deaths, cases-per-million, vaccination coverage
Power BI	Global Overview and Vaccination Analysis pages
Executive summary	One-page PDF with verified findings and limitations

Deliverable	Expected output
Documentation	README updated with methods, visuals and findings

Data source: Our World in Data COVID-19 dataset — https://github.com/owid/covid-19-data

Publication note: Mark visuals as completed only after verifying them in Power BI. Add final calculated values and screenshots before sharing the report.